MASHUK ELAHI

Marine & Offshore Electrical Engineer | Hybrid Power, Vessel Electrification & Engineering Digitalisation

Singapore

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PROFESSIONAL SUMMARY

Marine electrical engineer with 16 years of experience delivering low-voltage / medium-voltage / high-voltage (LV/MV/HV) power systems for offshore assets, vessel newbuilds, offshore-wind applications and hybrid-electric vessels. Specialised in power distribution, class-compliant electrical design, MV 11 kV protection, battery energy storage system (BESS) integration, power-management systems and commissioning through handover. Delivered first-pass approvals across ABS, BV, LR and DNV projects, including a hybrid-power architecture achieving up to 50% fuel reduction. Targeting senior marine, offshore, vessel-electrification and hybrid-power roles across Singapore, Europe, the UAE and APAC.

CORE COMPETENCIES

- LV/MV/HV Power Distribution & Class-Compliant Design
- MV 11 kV Protection & IEC 61850 Systems
- BESS, BMS & Hybrid Propulsion Integration
- Power-Management Systems & Shaft Generators
- SLDs, Load Lists & Cable Schedules
- ABS, BV, LR & DNV Class Approvals
- IEC 60092, IEC 61892, IEC 60364 & SOLAS
- ABB 800xA, FAT, SAT & FMEA
- Energisation, Sea Trials, Sail-Away & Handover
- Newbuild, FPSO, Jack-Up Rig & Offshore-Wind Substation Work

PROFESSIONAL EXPERIENCE

Electrical Engineer | Hybrid Marine & Vessel Electrification

Strategic Marine Pte. Ltd., Singapore | Jul 2024 - Present

- Designed LV power distribution for a multi-generator power system with three prime-power generator sets totalling 450 ekW; delivered single-line diagrams (SLDs), load lists and cable schedules, securing first-pass approval from four independent certification bodies with zero rework.
- Integrated a 72 kWh lithium-ion BESS, battery management system (BMS) and DC/DC inverters into a hybrid power architecture, reducing fuel draw by up to 50% during low-load operation.
- Led BMS-to-power-management-system interface work and Factory Acceptance Testing (FAT) through to class-compliant handover.
- Automated single-line diagram generation, load lists, cable schedules and alarm-list production with Python, AI agents and AutoCAD Electrical scripting, keeping drawings and schedules consistent across revisions.

Senior Project Engineer | Critical-Power & Prefabrication (Transferable)

Exyte, Singapore | Feb 2023 - Jul 2024

- Led 300 mm semiconductor cleanroom critical-power hook-up, energisation and Functional Performance Testing, achieving zero audit rework and first-pass PLC/SCADA interlock validation.

- Introduced prefabricated electrical power skids, reducing on-site installation labour by 50% - directly transferable to modular vessel and offshore power builds.

E&I Engineer | Renewable, Cogeneration & Hazardous Areas

Phoenix Mecano S.E.A. Pte. Ltd., Singapore | Jan 2021 - Jan 2023

- Delivered Malaysia's first solar-wind autonomous remote-power installation using a 24 V DC solar-battery bus, eliminating routine diesel-generator use and enabling remote/unmanned operation.
- Directed the E&I scope for an onshore cogeneration plant (transformers, MV/LV switchgear, 400 kVA UPS, SCADA integration) and designed IECEx/ATEX hazardous-area systems.

E&I Superintendent | Offshore Power Systems & Commissioning

Keppel FELS / Keppel Offshore & Marine, Singapore | Nov 2012 - Dec 2020

- Led commissioning of MV 11 kV switchgear and ABB 800xA SCADA-based fire-and-gas systems, maintaining MV/LV power reliability through final handover with crews of up to 100 personnel.
- Energised an offshore-wind substation electrical system to full IEC 61850 protection-relay conformity, achieving zero punch-list items at handover.
- Authored 30 E&I risk assessments and mentored 15 engineers through Innovation Quality Circles.

Assistant Electrical Engineer | Vessel Newbuild Electrical

Ananda Shipyard & Slipways Ltd., Bangladesh | Sep 2009 - Oct 2012

- Supported electrical installation, testing and commissioning activities on vessel newbuild projects, assisting senior engineers with equipment energisation and documentation.

SELECTED ACHIEVEMENTS

- Secured first-pass approval from four independent certification bodies with zero rework on a 450 ekW multi-generator LV distribution design.
- Reduced fuel draw by up to 50% at low load by integrating a 72 kWh BESS into a hybrid-electric vessel power architecture.
- Energised an offshore-wind substation to full IEC 61850 protection-relay conformity with zero punch-list items at handover.
- Led MV 11 kV switchgear and ABB 800xA fire-and-gas commissioning through handover with crews of up to 100 personnel.
- Delivered Malaysia's first solar-wind autonomous remote-power installation, eliminating routine diesel-generator use.

EDUCATION

- Master's Degree in Continuing Education in Sustainable Electricity Generation (UCAM, 60 ECTS, título propio) — UCAM Universidad Católica de Murcia, Spain — Apr 2026
- MSc Digital Transformation — London Metropolitan University, UK — Apr 2026
- Master of Business Administration (MBA) — University of Sunderland, UK — Jun 2025
- BSc Electrical and Electronic Engineering — Bangladesh University — Jul 2008
- Diploma in Electronics Engineering — Jessore Polytechnic Institute — Jan 2004

CERTIFICATIONS

- Certified Data Centre Professional (CDCP), EPI — 2026
- ISO 50001:2018 Energy Management Lead Auditor, GAQM / IGC
- ISO 9001:2015 Quality Management Lead Auditor, GAQM / IGC

- IECEx Ex 001: Basic Principles of Protection in Explosive Atmospheres — Valid through Oct 2027
- ABB Power System / LV Switchboard Training
- Fluke DSX2-8000 Cable Analyzer / Copper Certifier Training

TECHNICAL TOOLS

ABB 800xA; PLC/SCADA systems; IEC 61850 protection relays; EPMS/SCADA; Fluke DSX2-8000 Cable Analyzer / Copper Certifier.

PROFESSIONAL DEVELOPMENT

Areas of active study and continuing development (not claimed as project-verified experience):

- IACS UR E26 and E27 (vessel cyber-resilience) - in progress
- IEC 62443 industrial / marine cybersecurity - professional development
- Shore power / cold-ironing systems - professional development
- DC / LVDC zonal distribution architectures - professional development
- IEC 61660 (DC short-circuit) and IEC 62619 (industrial Li-ion safety) - in progress